SCBA

Personal Protective Equipment

any equipment which is intended to be worn or held by a person at work and which protects him against one or more risks to health or safety and any additional accessory designed to meet that objective.

Level A ensemble

- Fully encapsulating garment that completely envelops wearer and respiratory protection, SCBA
- Highest level of protection for skin, eyes, and lungs
- Effective against vapors, gases, mists, dusts
- NFPA 1991, *Standard on Vapor-Protective Ensembles for Hazardous Materials Emergencies*
- Category III, EN340: Type 1

Level B ensemble

- Multi-piece chemical-protective clothing, boots, gloves, and SCBA
- High level of respiratory protection but less skin protection
- Gloves and boots depend on physical and chemical properties of the chemical.
- Very common; often chosen for versatility
- NFPA 1992, *Standard on Liquid Splash-Protective Ensembles and Clothing for Hazardous Materials Emergencies*
- Category III, EN340: Type 3

Level C ensemble

- Used when airborne contaminant is known, its concentration is measured and criteria for using air-purifying respirator (APR) are met
- Worn with APR or powered air-purifying respirator (PAPR)
- Standard work clothing, chemical-protective clothing, chemical-resistant gloves, respiratory protection other than SCBA or SAR
- Used when significant skin and eye exposure is unlikely
- Category III, EN340: Type 3 or Type 4

Respiratory Protection – Type of respirator

Air Purifying

Half masks

Maintenance-free

Low maintenance

Full Facepiece

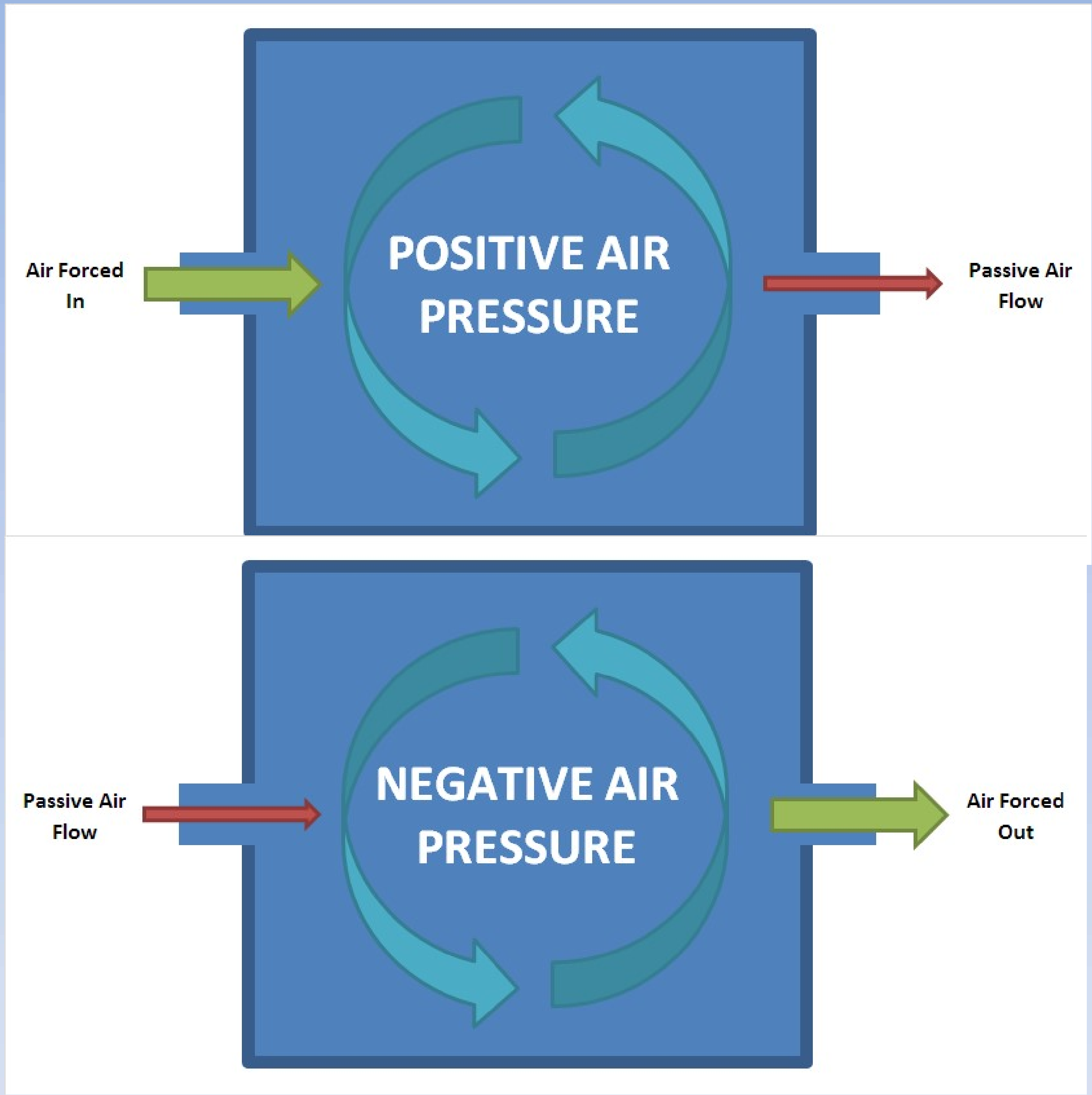
Powered Air Purifying Respirator (PAPR)

Supplied Air Respirators

Supplied Air Half and full facepiece, hoods or helmets

Self Contained Breathing Apparatus (SCBA)

Negative Pressure
Positive Pressure


Air Forced In
POSITIVE AIR PRESSURE
Passive Air Flow
Passive Air Flow
NEGATIVE AIR PRESSURE
Air Forced Out

Introduction

There are many jobs happening in environments that potentially contain toxics, but very often

- The type or the concentration of the contaminant is unknown
- The contaminant may suddenly appear or its concentration vary over time
- The contaminant is known to be very dangerous

These specific environments require a full respiratory insulation of the worker from the working area.

To ensure the product you choose is properly protecting you, it is key to understand the different categories of isolating devices to ensure you make the right choice when going to work everyday.

SCBA

What is SCBA

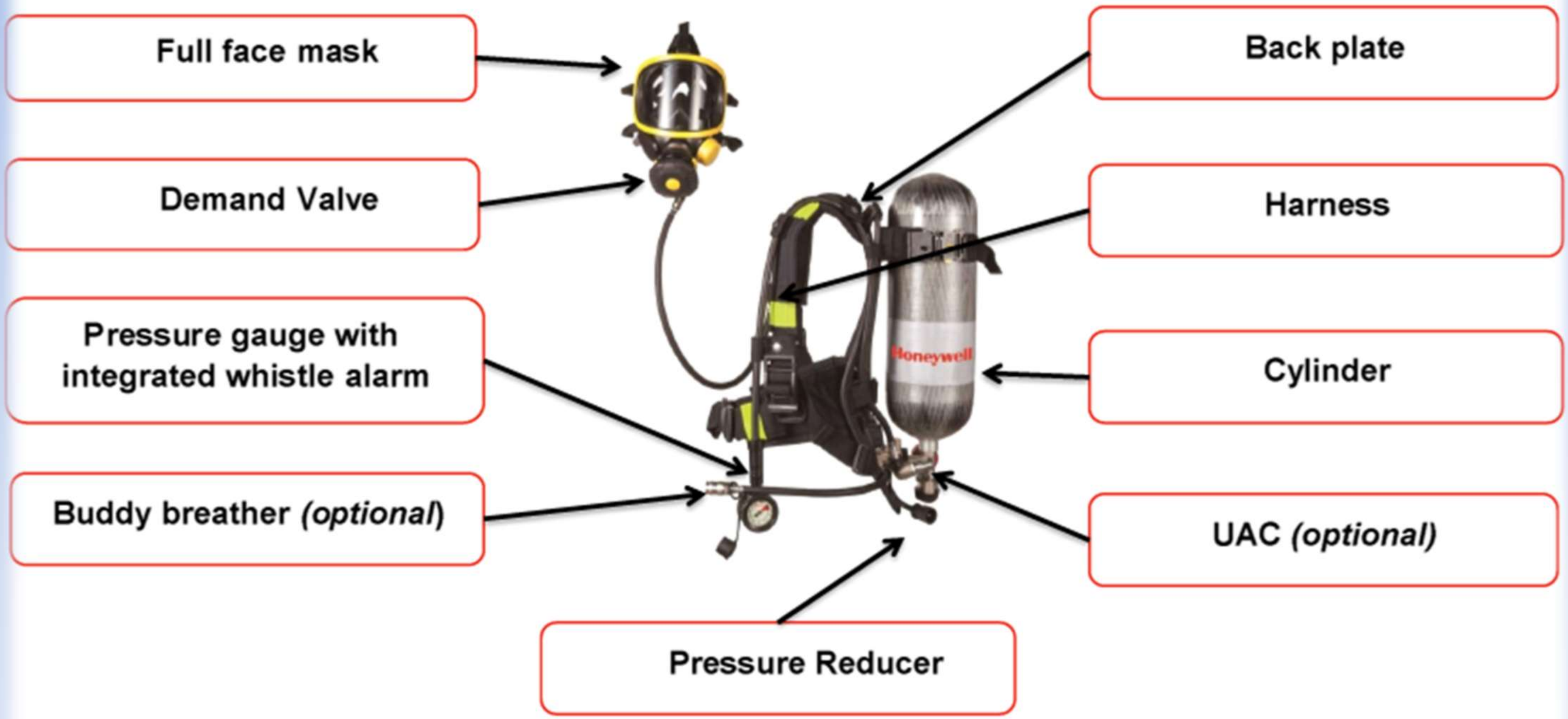
SCBA = Self Contained Breathing Apparatus

A device that provides autonomous breathable air in an IDLH (Immediate Danger to Life and Health) atmosphere (Toxic, Oxygen<17%,…) or any environment that is unknown or has unknown level of contaminants.

Target Verticals for SCBA

- Petrochemical Oil & Gas
- Fire Services (Government Fire Brigade or Volunteer Fire Fighting Services)
- Chemical (all confined space)
- Utilities (water clean & dirty)
- Nuclear
- Shipping (repair, transport)
- Heavy industry (fabrication, welding)
- Food industry
- Construction
- Military & Police

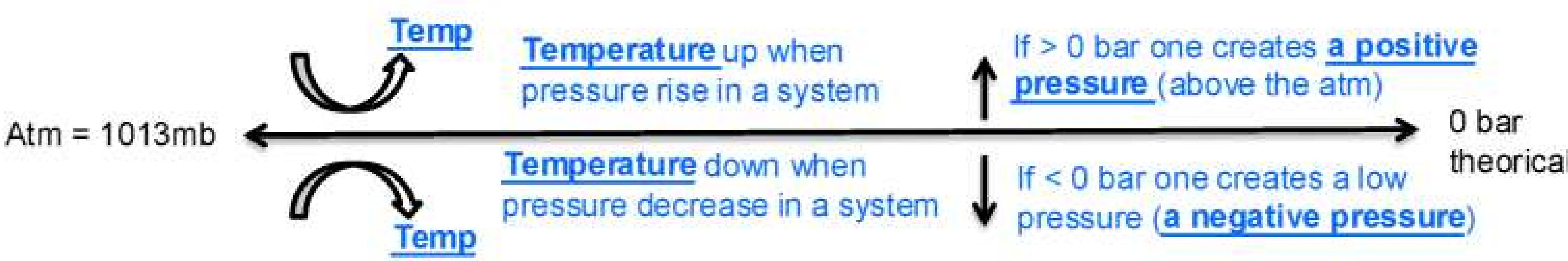
Main Components of SCBA



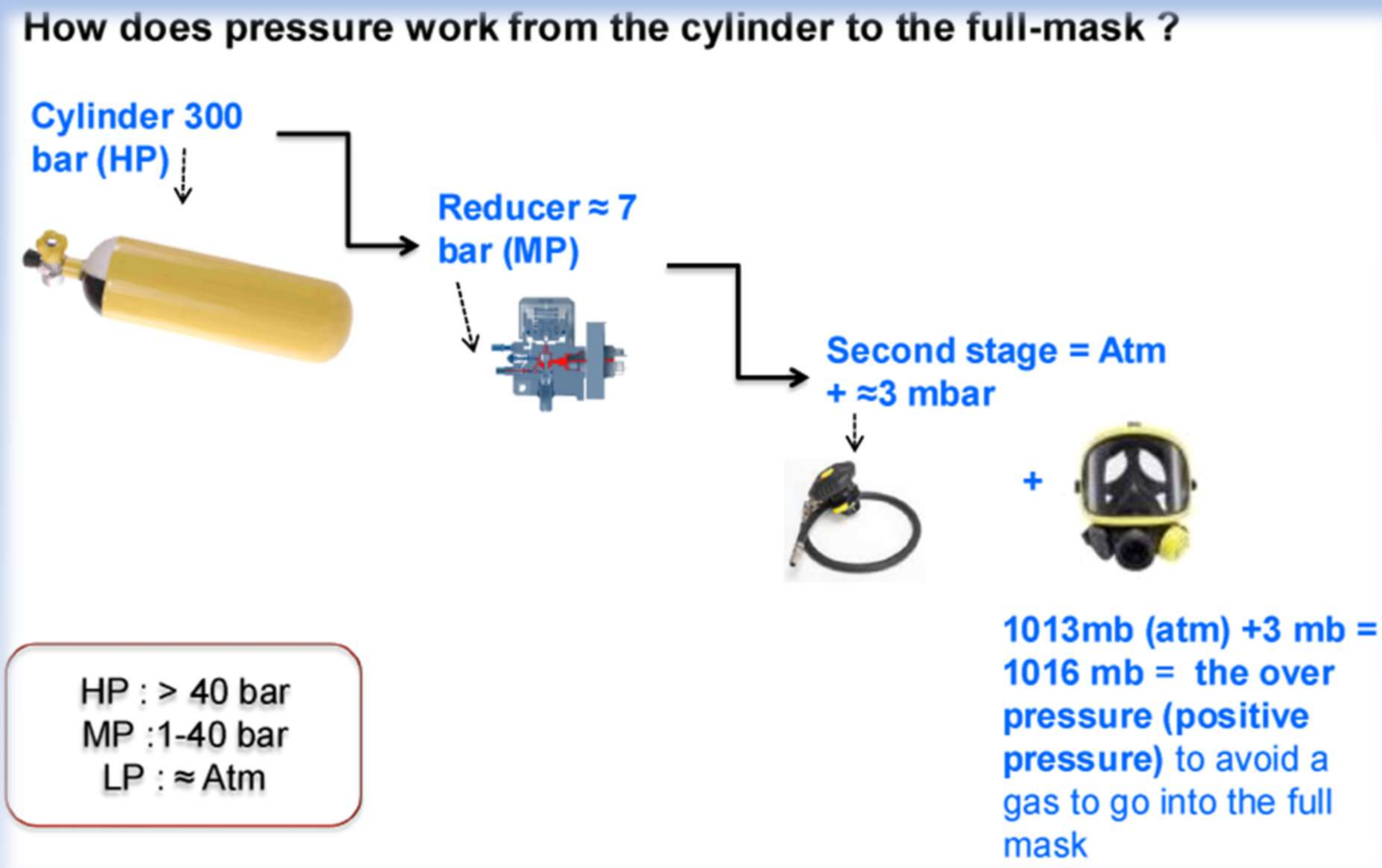
Principal of Positive Pressure SCBA

What is a pressure?

- Pressure is the effect of a force applied to a surface. As a weight by cm^2.
- Pressure unit can be expressed in 6 different ways. Pascal is the international unit. Bar is the generally use by every European version product.
- 1 bar = 1kg /cm^2 300 bar = 300kg / cm^2
- **Atmospheric pressure (atm) = 1,013 mbar = the "O" bar of the cylinder that comes out of the air compressor**



Principal of Positive Pressure SCBA



EN Regulation for SCBA

SCBA requirements are regulated by **EN 137: 2006** - Respiratory protective devices. Self-contained open-circuit compressed air breathing apparatus with full face mask. Requirements, testing, marking

5 Classification

Self-contained open-circuit compressed air breathing apparatus are classified in types as follows:

— Type 1: apparatus for industrial use;

— Type 2: apparatus for fire fighting.

Facepieces requirements are regulated by **EN 136: 1998** - Respiratory protective devices. Full face masks. Requirements, testing, marking

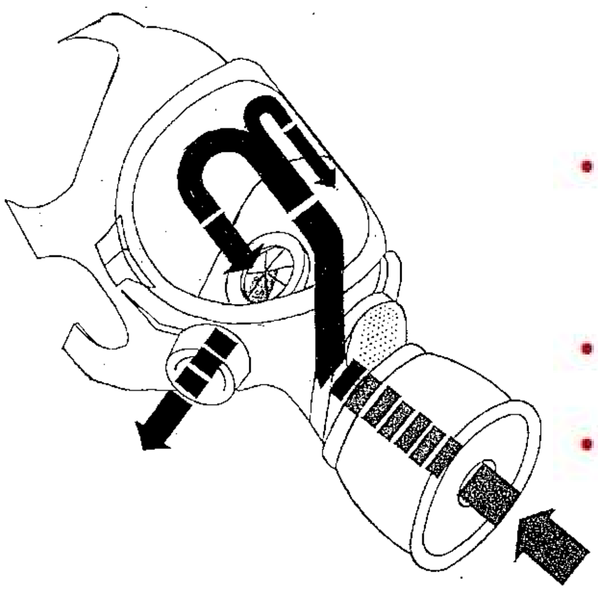
Cylinder valves requirements are regulated by EN 144:

- **EN 144-1: 2000** - Respiratory protective devices. Gas cylinder valves. Thread connections for insert connector
- **EN 144-2: 1999** - Respiratory protective devices. Gas cylinder valves. Outlet connections

Air quality for cylinders is regulated by **EN 12021: 2014** - Respiratory protective devices. Compressed air for breathing apparatus

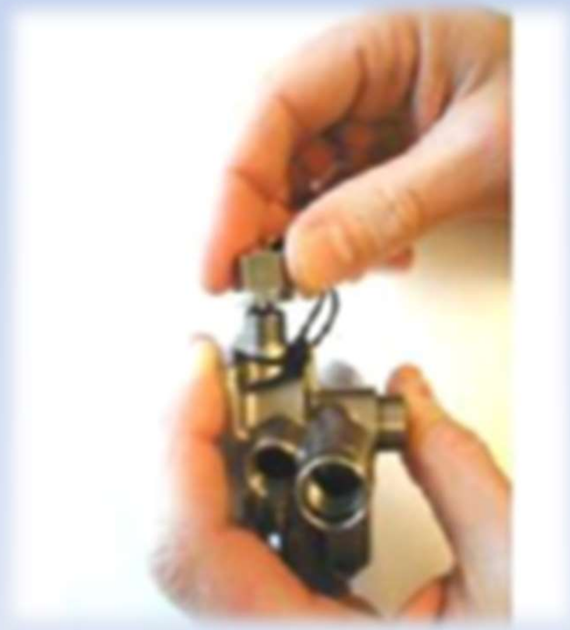
Pano Mask

- Certified to EN136, the lens are made of Polycarbonate class 3 with anti-scratch treatment, panoramic field of vision (96% of natural overlapped field of vision) and skirt in EPDM offering high comfort to the user
- A positive pressure facepiece with a quick connection to lung demand valve called 'Air-Klic'
- Equipped with speech diaphragm offering a good communication between users. This speech diaphragm is designed to be used with radio communication system.
- A cotton neck strap allows placing of the facepiece on the chest when not used with SCBA
- Enduring 850℃ temperature
- Weight 560 Grams

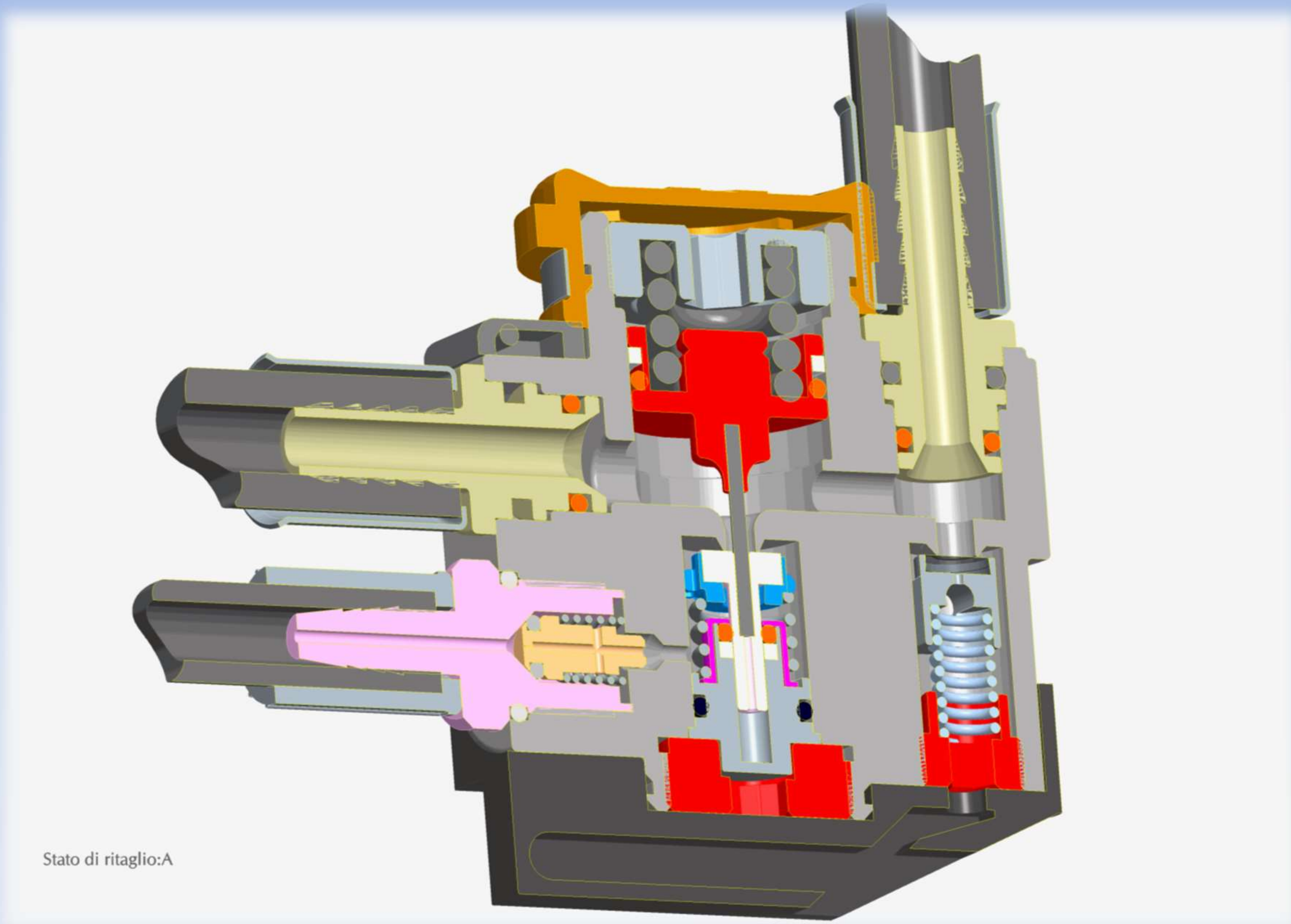


Pressure Reducer

- Piston valve to deliver air stability and reliability
- Maintenance friendly
- Two medium pressure output
- The medium pressure stage safety valve is adjustable but protected by security device to avoid any hazardous changing.



Pressure reducer, 3D



Lung Demand Valve

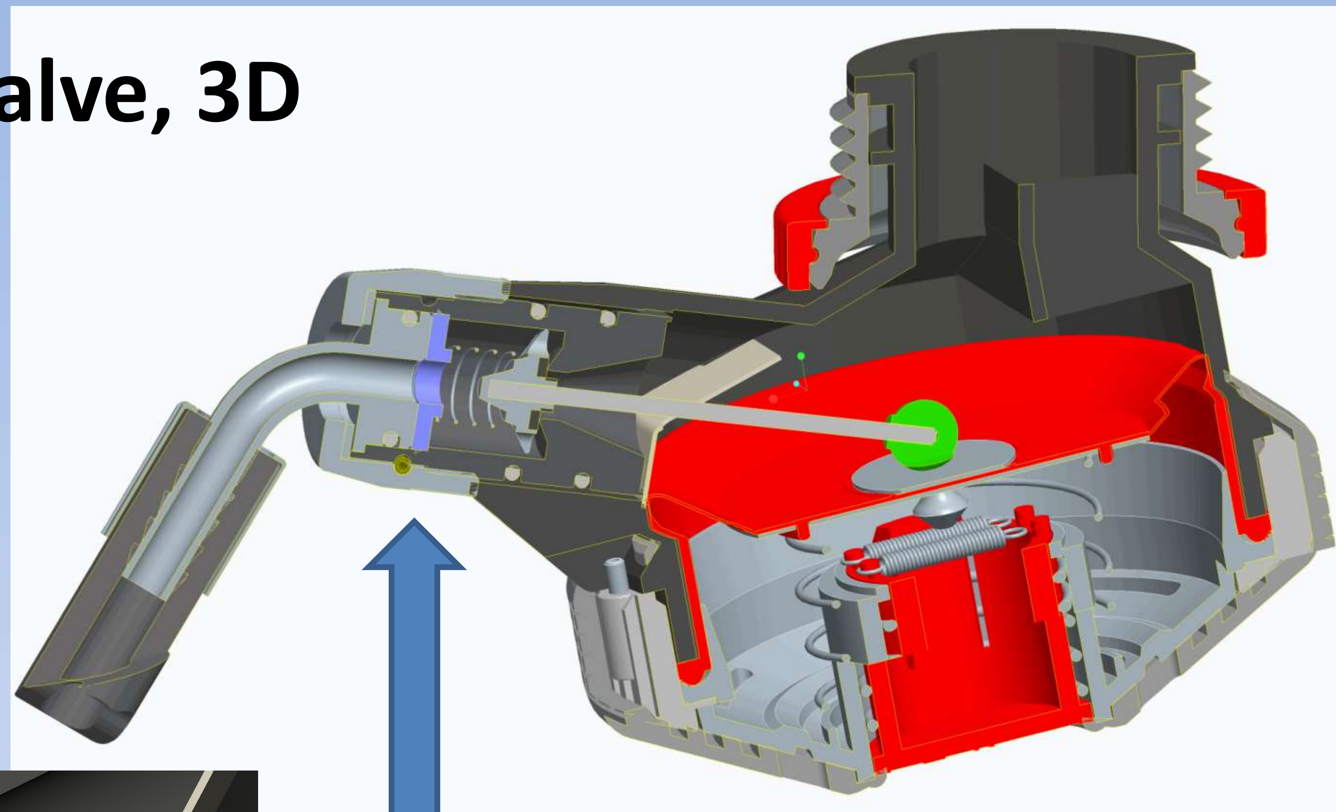
- The Zenith lung demand valve is positive pressure and automatic activation. When not connected to facepiece, no air flow. The lung demand valve is activated automatically when connected to facepiece

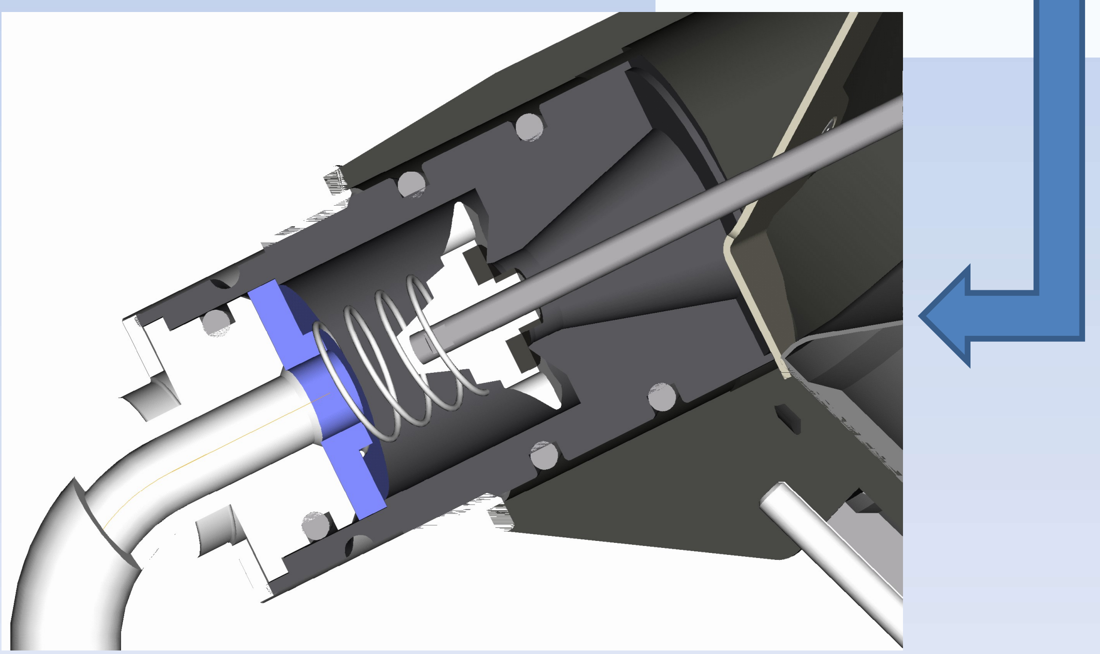
- The demand valve is equipped by a by-pass button. The flow in by-pass mode is 450 l/min.

- The demand valve fixation on the hose allows its rotation in two axes allowing easy hose positioning without any discomfort.

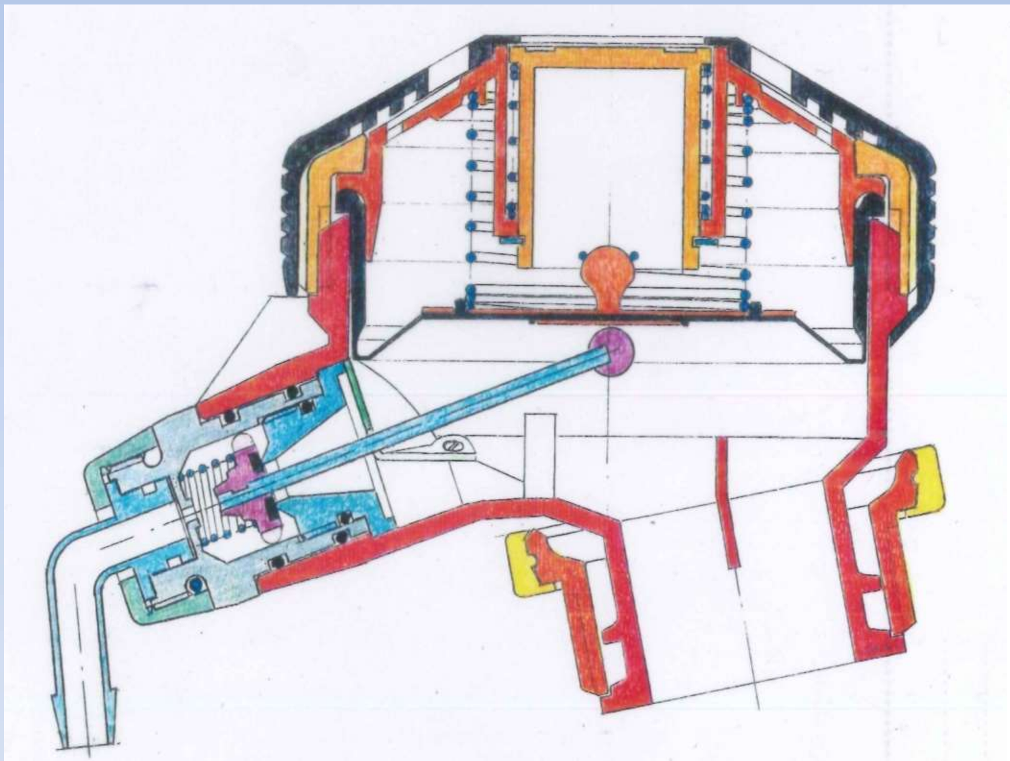
- Quick release from facepiece by pressing two large buttons.

Lung Demand Valve, 3D

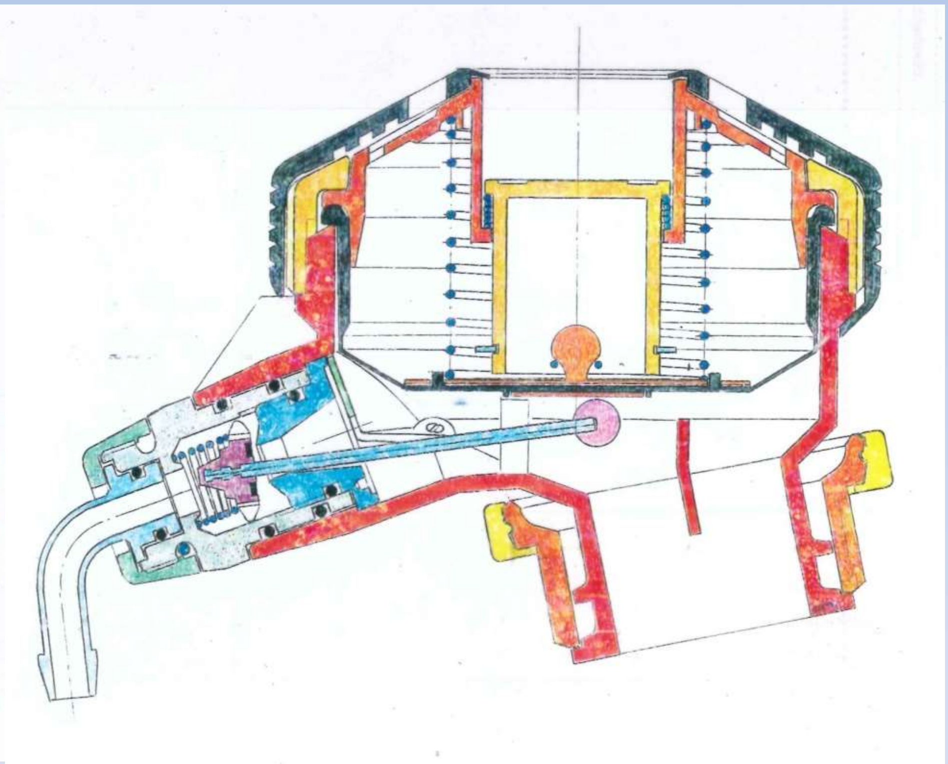




Not Working



Working



Mechanical gauge & alarm

- T8000 is equipped with mechanical gauge certified according to EN837-1 with precision class 1.6
- The alarm whistle is integrated in the handle to be closer to user ear
- The whistle output is protected by metal grid to avoid any obstruction
- Dual units gauge (Bar & Mpa)
- Operating characteristics:

- triggering pressure: 55 +/- 5 Bar
- whistle operates until the cylinder is totally empty
- air consumption less than 5 L/min
- audible sound: 90 dB
- audible frequency: > 2000 Hz

Backplate & Harnesses

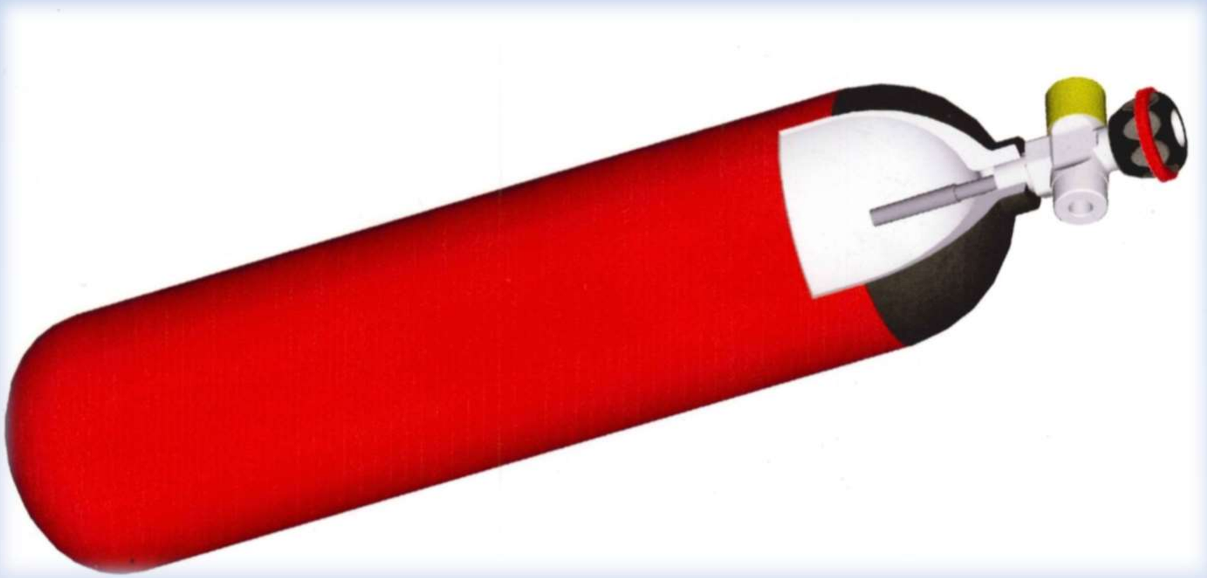
- Ergonomic designed for professional firefighting, compliant to EN137:2006 type 2 standard
- The backplate and its straps are resistant to radiant heat and self-extinguishing (flame engulfment)
- Comfortable harness with padding and with high visibility straps on shoulder straps and belt allows the users engage the mission in safety conditions
- Configured with cylinder size from 4.7L to 9L cylinder

Cylinder

- Light weight carbon composite cylinder with aluminium liner
- 15 years shelf life
- EN12245 Approval
- Equipped with pressure gauge, easy to observe remaining pressure without opening cylinder valve and without connect to pressure reducer
- Right hand wheel design; easy to control
- Standard configuration with rubber cap and rubber ring for more safety

Air properties for SCBA

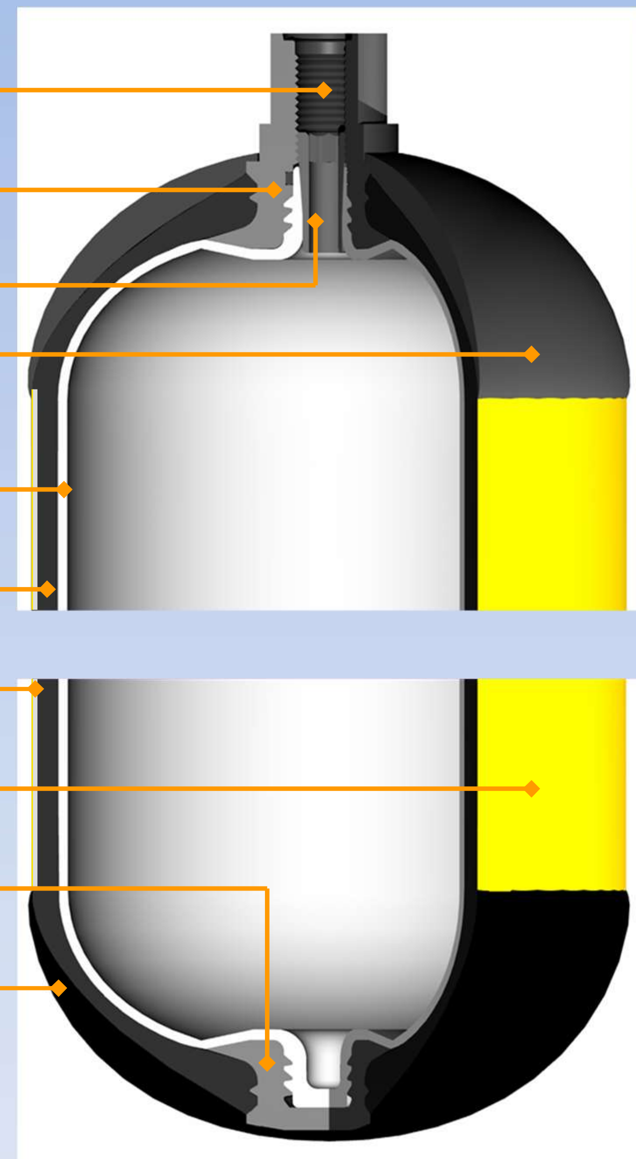
- $O_2 > 18\%$
- mineral oil < 0,3mg/mc - smell threshold
- vapor < 50mg/mc PER P < 300 bar
- Vapor < 35mg/mc PER P > 200 bar
- Toxics not present)



Air properties TEST

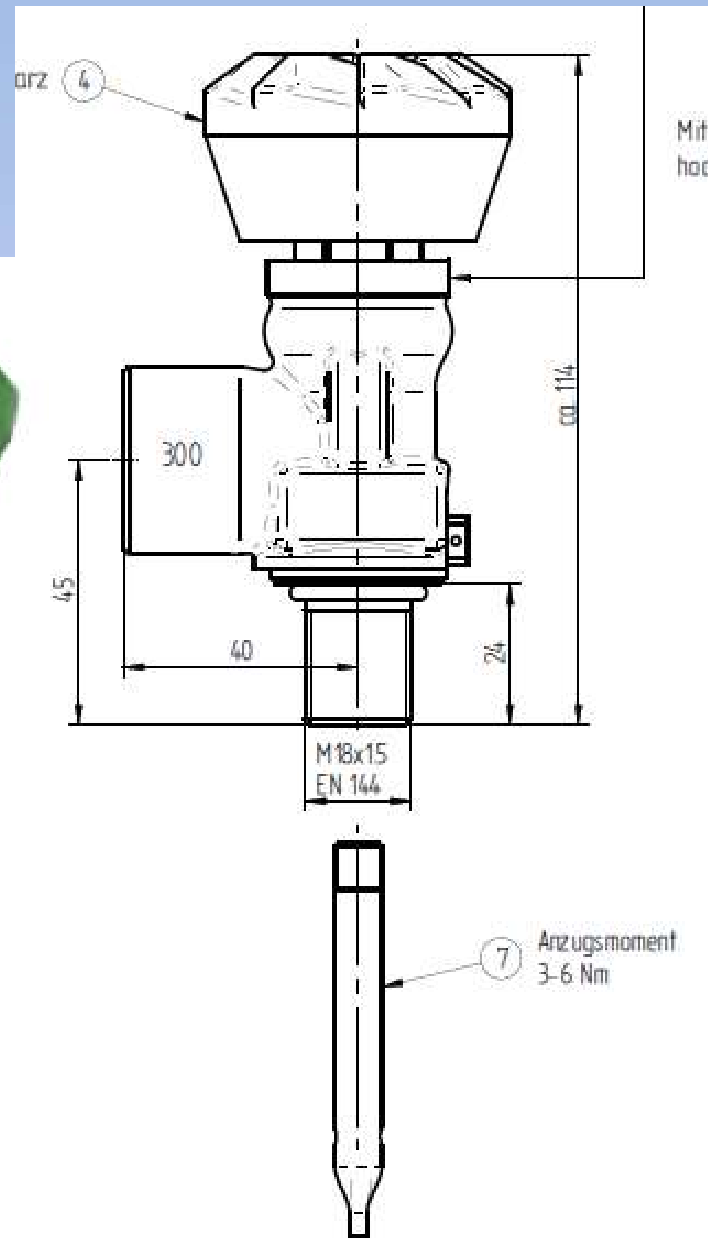
Composite Cylinder for SCBA

- M18 x 1,5 (valve thread)
- O-Ring (metal insert)
- Anello di pressione
- antishock protection
- inner bottle in plastic material)
- Carbon fibers)
- Fibre di vetro impregnate in resina epossidica
- Rivestimento di protezione in PVC termoplastico
- lower support
- antishock protection





Cylinder Valve



Excess flow Valve

Federaziun svizra dals pumpiers

Swiss Firefighters Association
Technical Breathing Apparatus Team

Highlights Technical Specification

- Weight with charged cylinder is 11.4~12 Kg
- Backplate Material:
  - Flame retardant Polyaramid-Fibre Glass backplate
  - Cushion in Nomex mixed with Kevlar harnesses
- Pressure reducer: 2 Outputs for separate 2 medium pressure hose
- Demand valve:
  - Maximum flow rate of by-pass mode is 450 l/min
  - Automatic activation when connect to facepiece
  - 360

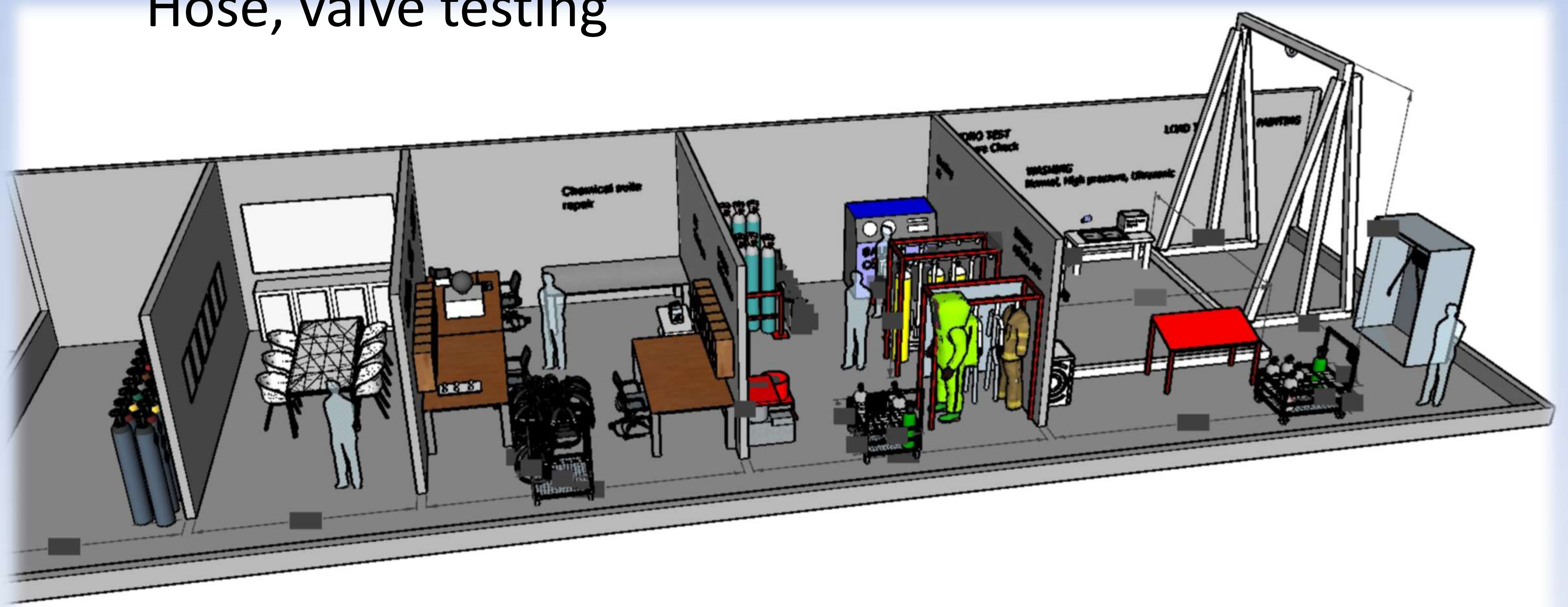
- Pano mask:
  - Weight 560 G.
  - Skirt and inner mask material is EPDM
  - Universal size fit for ASEAN face
- Pressure gauge with integrated whistle alarm

EN 137 type 1 vs type 2

Requirement	EN Type 1	EN Type 2
Resistance to radiant heat	N/A	8,0 kW/m2 , at 175 mm
Flame engulfment	N/A	90° ± 5° , 15min; then 950± 50 ° , 10s
Flammability Straps and buckles	800 ± 50 °, 12 ± **0.5s**	800 ± 50 °, 12 ± **0.5s**
Flammability breathing hose (s) medium pressure tube (s) and the lung governed demand valve	950± 50 °, 5± 0.5 s	950± 50 °, 5± 0.5 s
Breathing performances	20 x 2 l/min	20 x 2 l/min

SCBA Maintenance workshop

Visual inspection, posi check
Cylinder hydraulic expansion test
Cleaning, Drying, label,
Refilling of gases
Hose, valve testing



SCBA Maintenance

FREQUENCY OF MAINTENANCE AND INSPECTION OPERATIONS

COMPONENTS	*Type of work required*	*Before use*	*After use*	*Every six months*	*Every year*
Mask	Clean and disinfect (See mask's instructions for use)		X		X(3)
Full SCBA	Cleaning		X		X(4)
	Test on bench		X(2)	X(1)	X
	User function check	X	X		

1) For frequently-used apparatus
2) After use in aggressive environments or extreme conditions
3) For reserve stocks
4) Not if the apparatus is hermetically packaged

SCBA Maintenance

COMPONENTS	*Type of work to be done by a specialised maintenance workshop*	*Every year*	*Every 2 years*	*Every 6 years*
Mask	Replace: - inhalation/exhalation valves - seals		X	X(3)
Demand valve	Demand valve replacement			X
Pressure reducer	Pressure reducer replacement			X
	Replacement of the high pressure connector seal	X		
Compressed air cylinder	Regular inspection and requalification by an approved body	Refer to and follow the national regulations applicable to compressed air reserves		
Cylinder valve	Replace: - seal - closure member	Once every 5 years or more often		
	Replace: - Break seal	At least once every 12 months		

Test possibilities

Masks / Demand valves / CPS

- leakage tests in pos. and neg. pressure
- Opening pressure
- Working pressure
- Activate pressure to 10 l/min
- Static positive pressure

Breathing Apparatus / SCUBA

- High pressure leakage
- Pressure gauge comparison
- Response pressure warning device
- Medium pressure static
- Medium pressure leakage
- Medium pressure dynamic to 10 l/min

Cylinder hydraulic expansion test control unit

to determine the residual volumetric expansion of the cylinder under test pressure control.

A water jacket containing the cylinder during the test. According to the enclosed dimension technical data

Thank you